

brand recognition afforded to the PS2. When it comes to the eighth-gen consoles, the recurrent theme is compromise. Sony and Microsoft had adopted a loss-leader strategy when it came to their seventh-gen consoles.

At launch, both the PS3 and Xbox 360 were remarkably powerful pieces of hardware. The ATI-designed Xenos GPU powering the Xbox 360 held its own against ATI's 2005 and 2006 PC flagships. The PS3's Cell processor shifted to a wide, parallel, multi-threaded design years ahead of the curve. Intel's very first consumer quad-core part, the Q6600, arrived months after the PS3 launched in November 2006. These were high-end machines, expensive to manufacture and sold at a loss, the idea being that console games sold at \$60 dollars a pop would help recoup losses and drive long term profits. The upshot to this was that both the PS3 and Xbox 360 stayed relevant for a remarkably long time. When it came to game design, seventh-gen console hardware limitations weren't a major constraint for multiplats until as late as 2011 and the likes of Battlefield 3.

In contrast, the Xbox One and PS4 were behind the curve right from the day of their announcement: the off-the-shelf parts powering the two consoles were more or less on par with a lower-midrange PC circa 2012. Because of a highly constrained hardware spec, the eighth-gen consoles hit a brick wall in terms of hardware constraints soon after launch. As early as 2014, titles like Assassin's Creed: Unity were heavily compromised experiences on console, with sub-native resolutions and framerates in the mid-20s. With so little headroom left on their current hardware, Sony's taken the radical step of outing a mid-cycle hard-

ware refresh with the soon-to-be announced PS4 Neo. But is it a big enough step forward? We think that, even if a Polaris-toting PS4 refresh takes centerstage at Sony's conference, PC's still your best bet for a gaming platform this year. Here's why:

This latest graphics beast from NVIDIA's stable is not only capable of bringing the greatest AAA titles down to its knees, but it's a herald of even greater things



PC = Scalability

Let's start with the the most obvious reason. At least in terms of hardware, PC is a far more open platform than the consoles. (Let's just leave aside the part where Windows 10 hijacks your internet connection to seed forced updates on other people's devices.) You can run Steam off anything from a Pentium 4 lying around in an e-waste dump to a 14-core Xeon E5-2697. It's possible to tailor your PC experience to precisely match your needs. Can't afford a console? Grab a 650 Ti secondhand off of OLX for Rs. 5,500, slot it into your old desktop, and you'll be running current AAA titles at 720p/30 FPS in no time. Looking for a literal console experience? Get yourself a mini-ITX case and pair a Haswell or Skylake i3 with the ever-reliable 750 Ti. Xbox 360 Controller? Check. 55-inch TV? Check. You're good to go. And of course, if you want to power a 4K triple monitor array, you've

got the freedom to do that. If your needs happen to change at any point of time, you can slot in and slot out any part of your PC to make it work for you. Because they're fixed platforms, consoles can't have this flexibility. If you're upgrading to the PS4 Neo, you're not just paying for a new GPU--you're buying a whole system.

...14/16nm GPUs are incoming ...

After 4 years stuck on the 28nm process, NVIDIA and AMD are finally bringing their 14/16nm Fin-FET products to the market this year. What exactly does this mean? The smaller the process node, the more transistors can be squeezed into a given unit of space. This means smaller, cooler, more power efficient GPUs

across the board. At the higher end, cards like the GTX 1070 can offer Titan X-level performance at one third the price and with significantly lower power and thermal envelopes.

For the mainstream market, the incredible power efficiency gains mean that 50W parts like AMD's Polaris 11 GPU can outperform the PS4 and Xbox One without even needing a power connector. This is big news for value-oriented gamers. AMD in particular is focusing on increasing their addressable market with higher-performing budget SKUs based on the 14nm process. With Polaris 10 and 11 products set to unleash in the market, it will soon become possible to build PCs that outperform the consoles by a significant margin, while costing less, and running cooler and quieter.

On the other hand, the jump to 14/16nm means that we're likely to jaw-dropping levels of performance in the enthusiast segment later in the year. The GTX 1080, out in the market already, is a herald of better things to come. The GP104 GPU at its heart is 30 percent faster than GM200, but at just half the die size. We've already seen the massive GP100 in limited form with the Tesla P100. A hypotheti-